AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 4 · TEACHER GUIDE

Python and Table Readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach one connected table-checking story.

Separate core schema and summary reasoning from optional file and recovery extensions.

Keep one habit: Use exact keys, types, units and counts; assess the selected mean task before feedback.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Specify a table **schema**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach Week 13 as a two-session investigation using fictional data. A fictional classroom logs the duration of each activity.

Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Look up the value associated with the key "used" in that record.

Python 3 editor with a fresh local practice **file**

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: File input and output

Teach Week 25 as a two-session investigation using fictional data. Use a teacher-provided local sandbox folder, never a real personal **file**. Create durations.txt with two lines: 12 and 8.

The goal is to read the text, convert it to integers and report the total. Keep the original **file** unchanged for this activity. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

It is text; conversion and validation are needed.

Python 3 editor with a fresh local practice **file**

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Exceptions and recovery

Teach Week 27 as a two-session investigation using fictional data. A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60.

Invalid numeric text should return a clear **status** rather than crash or pretend that the duration was zero. Test both the successful and failing **path**s.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

ValueError.

Python 3 editor with a fresh local practice **file**

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Opening retrieval

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a task from the meaning of its target

Reference example, not the only acceptable wording: Route **type** is classification because 1 and 2 encode categories. Whole-minute duration is regression because numerical differences measure time.

Exceeding 10 minutes is binary classification because the target is yes or no.

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish **fitting** a rule from applying a fixed rule

Reference example, not the only acceptable wording: The **input** is width x and the label is the observed category 0 or 1. Candidates are 2, 4 and 6.

Selection uses mistakes on the four **training** cards only; held out labels must not guide it.

End goal

By the end, I can explain python and table readiness using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Case exposure check: this source example appears in ai-2/7, ai-4/4. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Classification or Regression?

The ML Lifecycle

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

Concept to explain

Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.

Term	Meaning
variable	name bound to a value
data type	kind of value and valid operations
dataframe	table represented in code
expression	code that computes a value

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach contracts, then summaries, then optional extensions

> Model one complete table before adding files and exception recovery.

Core schema and exact-type reasoning

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries.

The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written **schema**.

Term	Meaning
Schema	Rules for the fields and values in records
Type	A category such as integer or text
Unit	The quantity scale attached to a number
Constraint	A rule a valid record must satisfy

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

Complete plant records and measured coverage

Inspect a small table before asking Python to compute an average.

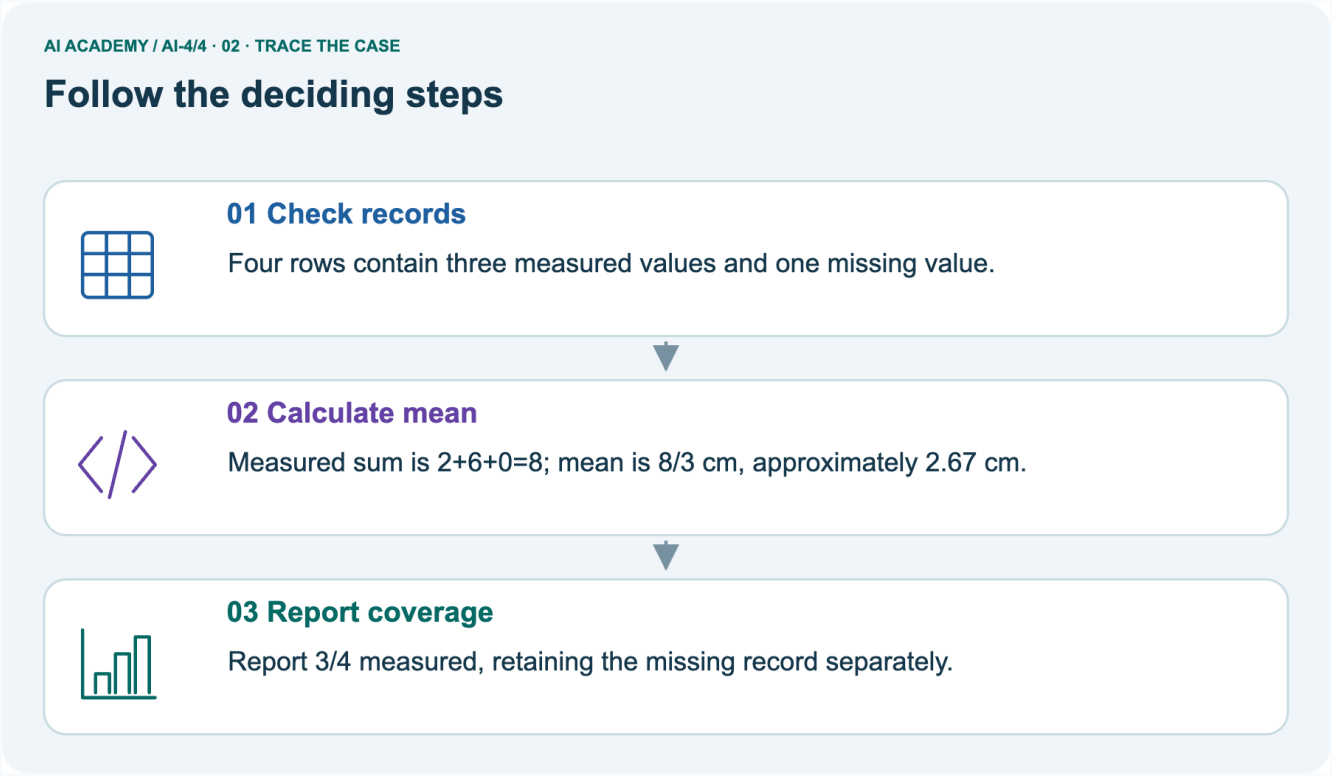
Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

This height table uses its own contract: heights are nonnegative measured numbers in cm or None for unmeasured. It is not the activity-duration schema. Exclude None explicitly, retain the observed zero and keep the original four-row denominator for coverage. The mean of available observations does not establish the mean of all four plants.

If no heights are measured, report mean unavailable and coverage 0/4; sum 0 alone does not imply measured zero height.

Trace the mean and limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check records	Four rows contain three measured values and one missing value.
2. Calculate mean	Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.
3. Report coverage	Report 3/4 measured, retaining the missing record separately.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

The measured-row mean may not represent the missing plants; investigate why those readings are absent.

Predict zero becoming unmeasured

Change: S changes from 0 to None.

Prediction: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

▼ Optional file-reading contract and checked outputs

Use a teacher-provided local sandbox folder, never a real personal **file**. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total.

Keep the original **file** unchanged for this activity.

The mode r means read. with closes the **file** when the block finishes, including when an error occurs. splitlines separates lines without retaining newline characters.

This example assumes each line is a valid integer text; an empty line or twelve raises ValueError during int conversion.

Values [0, 3, 7], total 10. Exit: It is a parse failure, not a measured zero. Flag the line and report the problem; silently substituting changes the **evidence**.

▼ Optional parser contract and checked statuses

A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear **status** rather than crash or pretend that the duration was zero.

Test both the successful and failing **path**s.

try covers the operation expected to fail; except ValueError handles that known conversion failure. Returning early ends the failing **path**. This contract expects a string: it does not claim to handle every possible object.

Spaces around an integer are accepted by int, and that behavior should be documented.

out of range/None, ok/5, invalid text/None. Exit: Accepted count 8, rejected count 2 and reasons; summaries apply to accepted data. Preserve raw **input** for authorized correction.

▼ Additional list-mean modelling

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Trace a short notebook state and repair two **type** or missing-field errors.

MODEL IDEA Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD widths = [4.2, 5.0, 6.1] stores three decimal measurements. mean_width = sum(widths) / len(widths) evaluates to 5.1.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 widths = [4.2, 5.0, 6.1] stores three decimal measurements.

STEP 2 mean_width = sum(widths) / len(widths) evaluates to 5.1.

STEP 3 Adding 'unknown' to the numeric list causes a **type** problem.

STEP 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Apply the connected idea: File input and output

Apply the connected idea: Exceptions and recovery

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

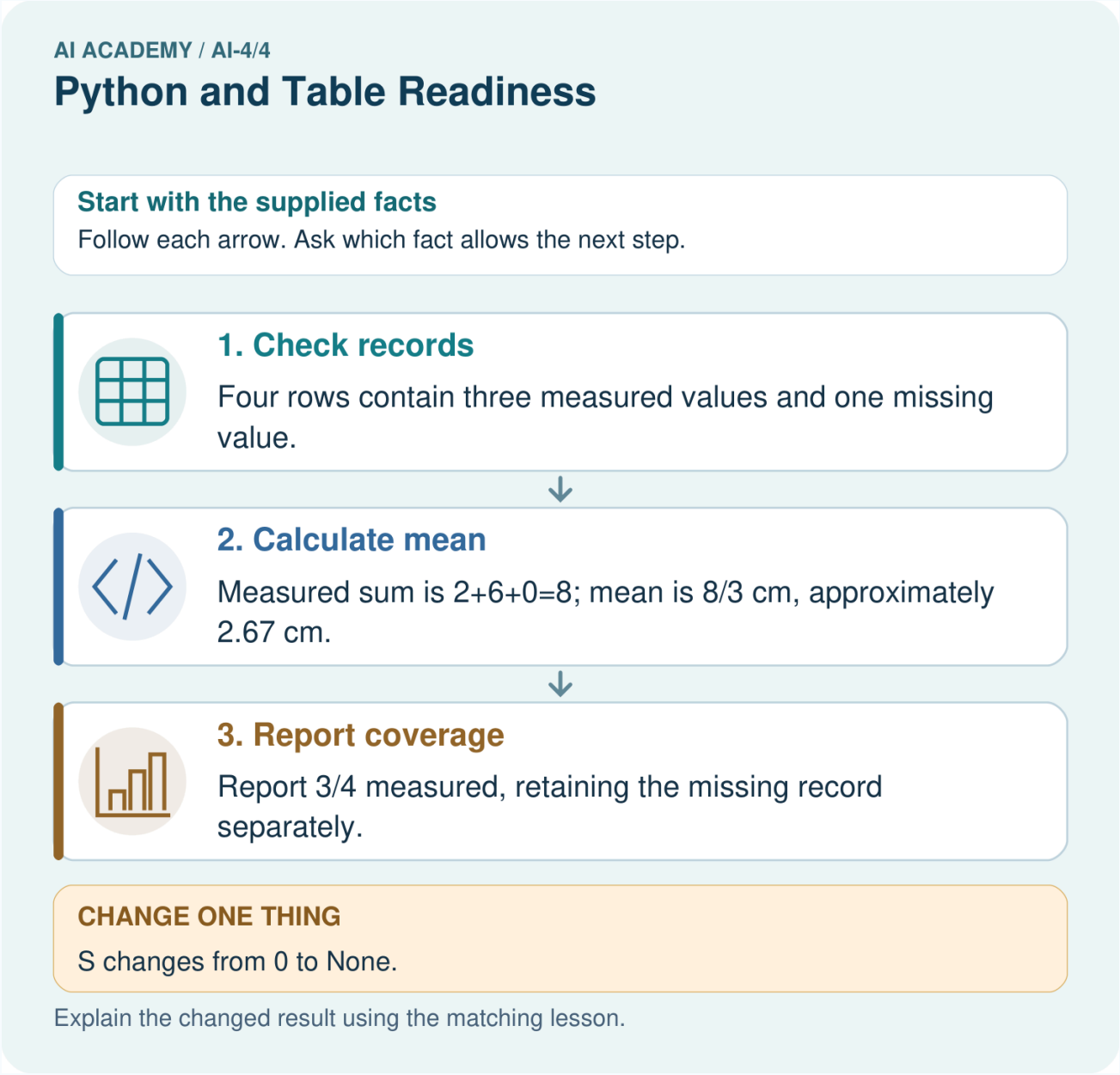
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Python and Table Readiness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Classify minutes values 0, 60, -1 and 61 under this **schema**.

Teacher check: 0 and 60 pass; -1 and 61 fail.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Rows columns and **schema**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Zero is a real recorded measurement here. Replacing missing with zero would invent **evidence** and lower the mean without justification. Changed-case check: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Removing a measured zero changes both the denominator and the coverage.

Changed-case reasoning: Mean becomes (2+6)/2=4 cm with 2/4 measured. Removing a measured zero changes both the denominator and the coverage.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information **path** or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Write the question to explain: Trace and handle an empty list.

Reveal after **prediction**s: Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

Use the earlier transfer question as additional practice. Reserve the supplied independent observed-values and missing-record task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

61 fails the range; "12" fails the integer **type**; 12 passes. Exit: No. The value has a permitted structure but still needs trustworthy collection **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.

Stage	Teacher action
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

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Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
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33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

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33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

▼ Extension teaching cycles

Connected teaching cycle: File input and output

Target: Read a small local data **file**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A sandbox **file** has lines 5, 0 and 2. What total does the worked reader print?

Teacher check: The parsed values sum to 5 + 0 + 2 = 7.

Individual check: For **File input** and **output**, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: **File** reading does not establish that the values are valid durations. After conversion, apply the **schema** and record any rejected lines. Do not upload personal records to an external tool for this exercise.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Connected teaching cycle: Exceptions and recovery

Target: Handle expected **exception**s specifically

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Compare parse_minutes("abc") and parse_minutes("70").

Teacher check: abc gives invalid text/None; 70 gives out of range/None. Both fail but for different reasons.

Individual check: For **Exception**s and **recovery**, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: **Recovery** must fit the task. A caller may ask for corrected **input** or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully **process**ed.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Notebook key: counts=[2,0,6], total=8, n=3, mean=8/3, observed=3. Minute-range filter retains A,B; positive-only filter A. 12+"8" raises TypeError; intentional conversion gives 20, subject to declared minute constraints. Missing minutes raises KeyError: obtain evidence, not

zero. Probes:12 passes,0 passes,missing needs correction,120 seconds documented as2 minutes.

Selected independent values3,7,14 sum24/count3 mean8; empty unavailable; fourth missing means available-case coverage3/4, not a known full mean. Optional file and parser keys remain in named extension references.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Notebook Detective: predict five **variable** values, trace two table filters, repair two errors, and rerun from a clean state.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A successful run proves the conclusion: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: define a tray label and integer count 0 through 20. Require tests at 0, 20, -1 and 21 and a text count.

A convenient shortcut preserves meaning: If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading.

Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: **File** reading does not establish that the values are valid durations. After conversion, apply the **schema** and record any rejected lines.

Do not upload personal records to an external tool for this exercise.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: sandbox values 2, 7, 1, 0 give count 4 and total 10.

An empty **file** yields count 0 and sum 0, which must be labelled no **observation**s.

A convenient shortcut preserves meaning: If Python cannot find durations.txt, it raises **File**NotFoundError. Check the current working folder and exact **file**name.

Do not switch to write mode to solve a missing **input**: opening an existing **file** with w would replace its contents.

Check these misconceptions

A successful run proves the conclusion: **Recovery** must fit the task. A caller may ask for corrected **input** or exclude the row with a recorded reason.

It must not use None as a number or claim that rejected records were successfully **process**ed.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: integer score strings 0 through 5. "5" ok/5, "6" out of range/None, "" invalid text/None, "0" ok/0.

A convenient shortcut preserves meaning: A bare except that returns 0 hides unrelated programming errors and corrupts missing values. Catch the expected **exception**, preserve an error **status** and keep the unknown value as None.

If **file** reading is added, handle **File**NotFoundError separately with an actionable message.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the supplied independent observed-values and missing-record task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR If notebook cells eventually run, execution order and **data type**s do not matter.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Supplied values are3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

Mean=(3+7+14)/3=8. An empty list has no observed mean; return an explicit unavailable result rather than divide by zero or invent0.

State the missing-value policy and report coverage3/4 separately if one of four records is missing.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 State a schema

List the required fields, **type**s and duration boundaries for the worked table.

activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Check both fields. activity must exist as nonempty text. minutes must be an exact integer and satisfy 0 <= minutes <= 60. Inclusive means equality is accepted at both ends.

Structure, **type** and range are separate checks; passing them does not prove the duration was measured accurately.

W2 Classify records

Classify minutes values 0, 60, -1 and 61 under this **schema**.

0 and 60 pass; -1 and 61 fail.

Apply the inclusive range to each integer. Both 0 and 60 are on allowed boundaries. -1 is below the minimum and 61 is above the maximum. Require the reason for each rejection, not only pass/fail labels.

W3 Repair a unit problem

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

Divide by 60 to record 3 minutes and retain the source value and conversion note.

W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

No. The value has a permitted structure but still needs trustworthy collection **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

Additional transfer practice question

Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.

Reasoned practice answer

Python indexes this list from zero: lengths[1] is 7, so selected is 7 and total is 10.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Index 1 always selects the first list item.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Sum=24, count=3, mean=8.0. For [] the mean is undefined; explicitly report missing/insufficient data before division. Returning zero invents a measurement and can distort later **prediction**s.

Test both populated and empty **input**s.

Explanations for the additional workbook examples

W1 State a schema

W2 Classify records

W3 Repair a unit problem

W4 Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count **type**. Accept alternatives meeting the **written rule**s.

W5 Separate validity and truth

Assess independent transfer

Selected case answer

Feedback that the learner uses

▼ Extension questions and checked keys

File input and output — W1 Trace types

Before and after int("8"), identify the **type**s and values.

Before: text "8". After: integer 8.

Before conversion, "8" is a string containing the character 8. int("8") returns the integer 8, which can participate in numeric arithmetic.

The conversion does not change the original **file** and does not establish that the number meets the duration **schema**.

File input and output — W2 Predict the total

A sandbox **file** has lines 5, 0 and 2. What total does the worked reader print?

The parsed values sum to 5 + 0 + 2 = 7.

File input and output — W3 Repair a path issue

The **file** is in the class-data folder but the script looks in its working folder. Give a safe repair.

Use the correct **path** to class-data/durations.txt or run from the intended folder. Keep read mode and do not overwrite the **input**.

File input and output — W4 Read independently

In a fresh sandbox **file** counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

Read counters.txt as UTF-8 text, convert three lines to integers; count 3 and total 10. Keep the raw **file** unchanged.

File input and output — W5 Handle malformed data

The **file** contains twelve on one line. Why should the program not silently turn it into zero?

It is a parse failure, not a measured zero. Flag the line and report the problem; silently substituting changes the **evidence**.

Exceptions and recovery — W1 Trace success

What does parse_minutes("0") return, and why is zero allowed?

A record with **status** ok and value 0; zero meets the inclusive range.

Exceptions and recovery — W2 Trace two failures

Compare parse_minutes("abc") and parse_minutes("70").

abc gives invalid text/None; 70 gives out of range/None. Both fail but for different reasons.

Exceptions and recovery — W3 Repair a handler

An except block returns 0 for any failure. State two problems and a repair.

It confuses failure with observed zero and may hide unrelated bugs. Catch ValueError, return an explicit **status** and None, then validate range.

Exceptions and recovery — W4 Build independently

Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record **status**/value.

4 and 0 give ok with 4 and 0; 11 and -1 give out of range/None; four gives invalid text/None. Use specific handling and explicit range checks.

Exceptions and recovery — W5 Choose recovery

A batch has 8 valid and 2 invalid lines. What should its report include?

Accepted count 8, rejected count 2 and reasons; summaries apply to accepted data. Preserve raw **input** for authorized correction.

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit **unit**s and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use a **schema** card beside each record. Extension: explain how allowing decimal durations would change both **type** and rounding rules.

Selected independent **evidence**: observed sum 24, count 3, mean 8. Empty collection means mean unavailable. A missing fourth record stays unknown; under an explicit available-case policy report mean 8 for the three observed records and coverage 3/4, without claiming the four-record mean. Fresh retry: observed values 2,4,9 and two missing records give sum 15, observed mean 5, coverage 3/5. Record help and prior exposure.

Delayed return

Suggested return points in this level: ai-4/5, ai-4/7, ai-4/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 5: Inspect Before Modeling. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Missing values and validation. Reuse this week's explicit contracts and **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

▼ **Extension support and reassessment**

Readiness and reassessment for File input and output

Provide the exact **file** and folder **path**. Extension: retain line numbers alongside parsed values to make corrections traceable.

Independent W4 **evidence**: Read counters.txt as UTF-8 text, convert three lines to integers; count 3 and total 10. Keep the raw **file** unchanged. Record whether the learner needed no help, a hint or explanation.

Fresh retry: sandbox values 2, 7, 1, 0 give count 4 and total 10. An empty **file** yields count 0 and sum 0, which must be labelled no **observation**s.

Readiness and reassessment for Exceptions and recovery

Use success and failure **path** cards. Extension: reject non-string **input** explicitly with a separate **status** before calling int.

Independent W4 **evidence**: 4 and 0 give ok with 4 and 0; 11 and -1 give out of range/None; four gives invalid text/None. Use specific handling and explicit range checks.

Record whether the learner needed no help, a hint or explanation. Fresh retry: integer score strings 0 through 5. "5" ok/5, "6" out of range/None, "" invalid text/None, "0" ok/0.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Specify a table **schema**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Classify minutes values 0, 60, -1 and 61 under this **schema**.

Reasoned practice answer: 0 and 60 pass; -1 and 61 fail.

Check every learner before discussion: Use the worked case for Rows columns and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: S changes from 0 to None. Require a **prediction** and a reason before comparing. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

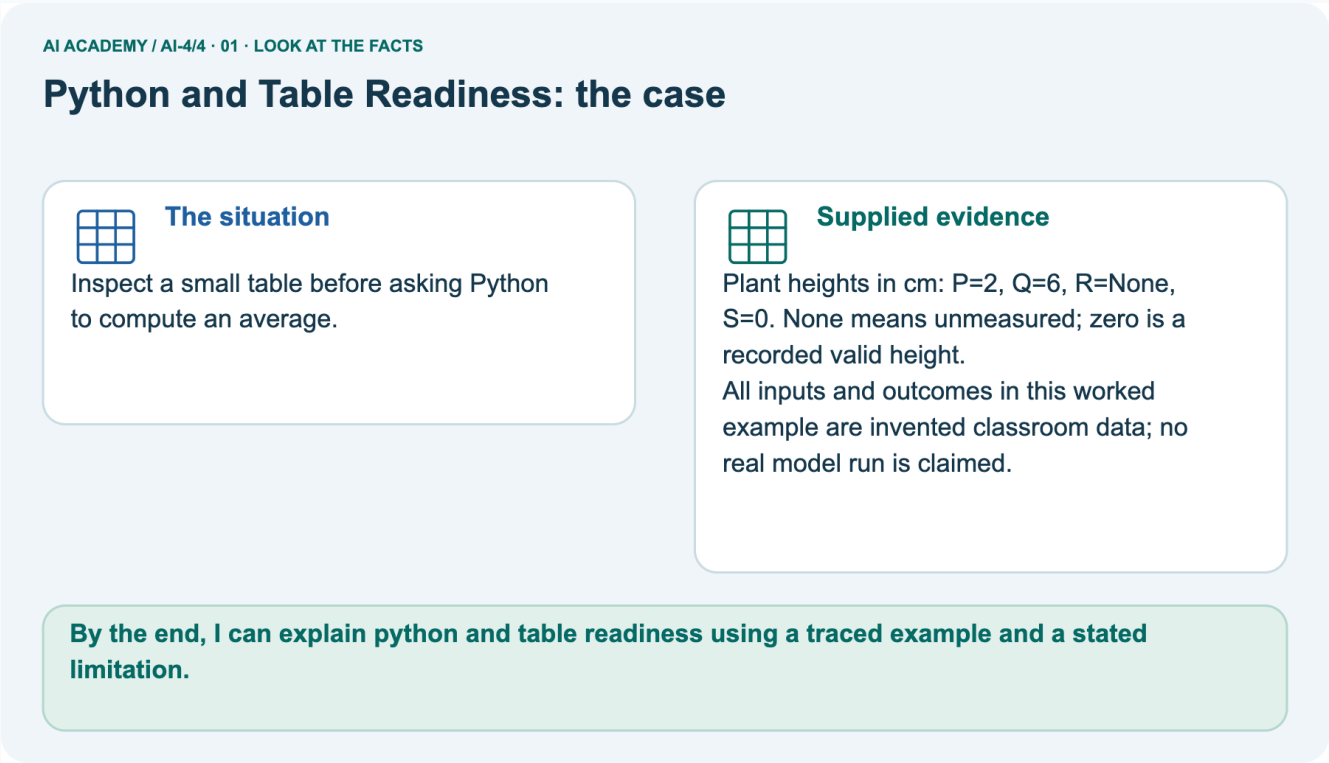
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Python and Table Readiness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

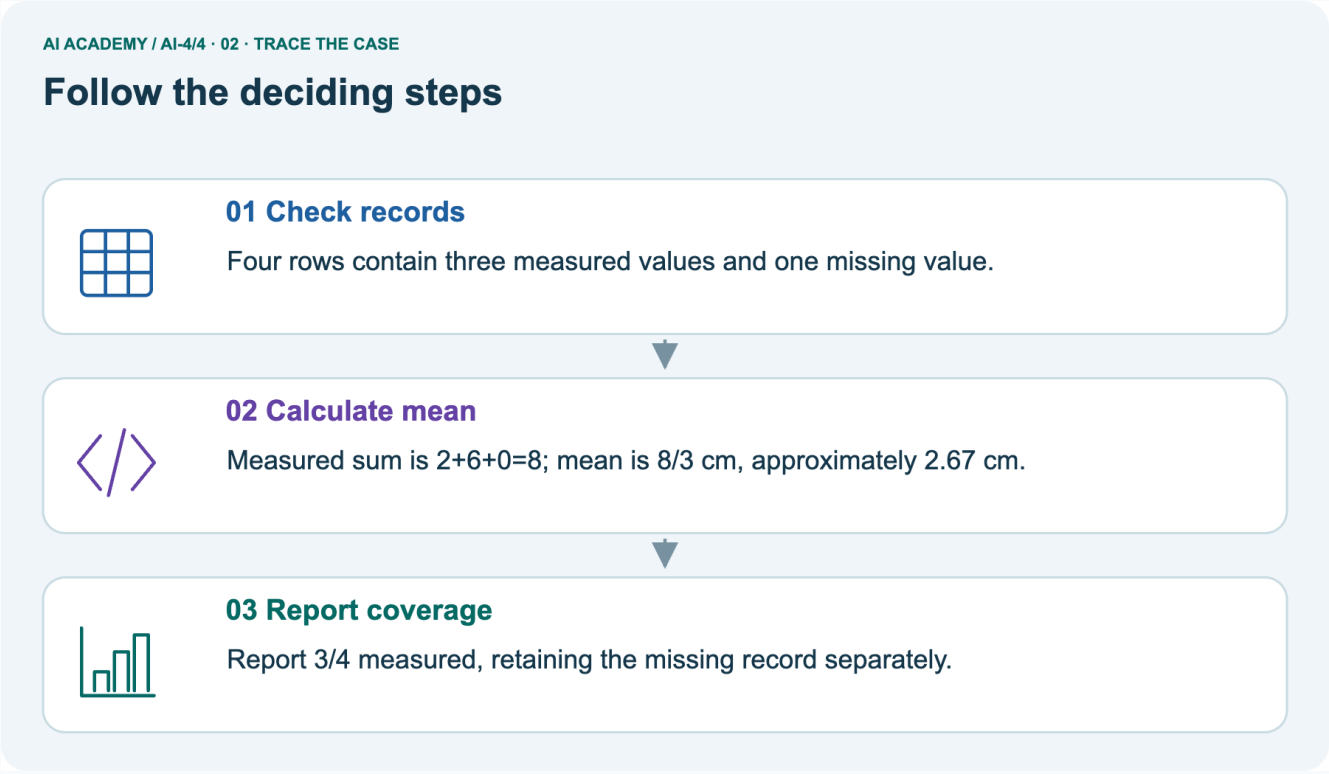
Inspect a small table before asking Python to compute an average. Supplied facts: Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



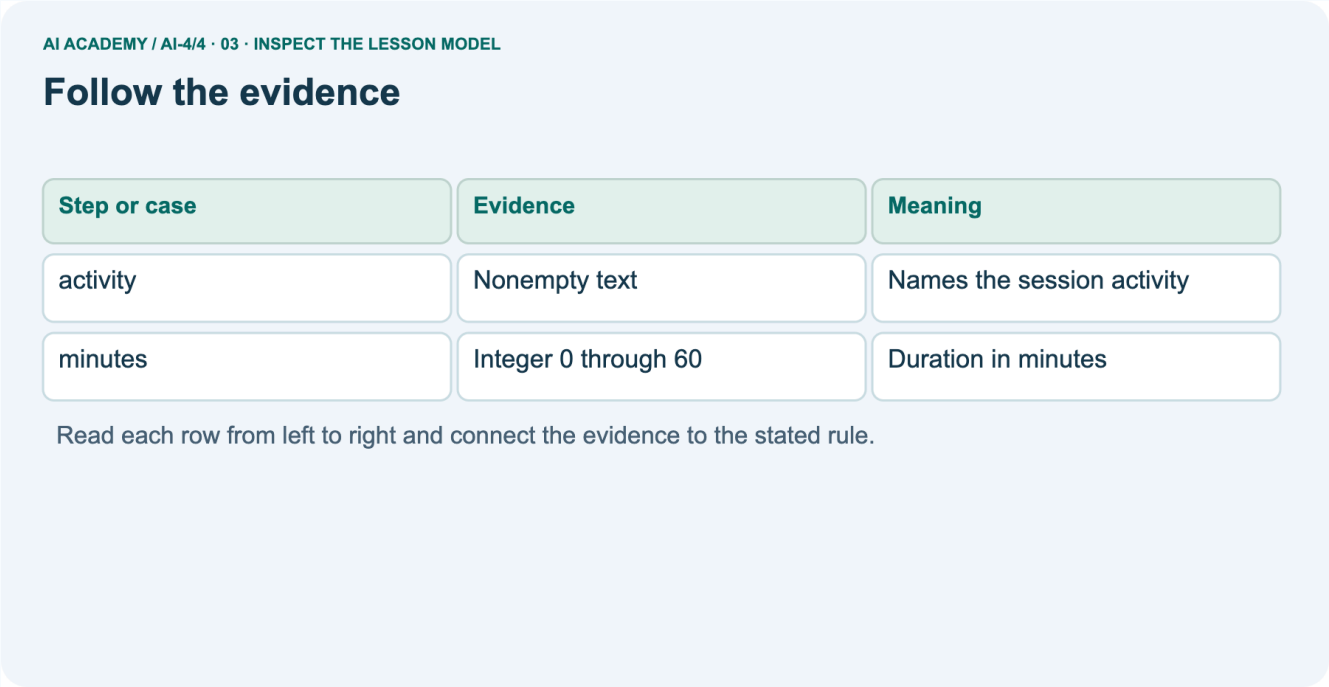
Follow the labelled picture with the matching explanation. Open the vector diagram

Check records: Four rows contain three measured values and one missing value. → Calculate mean: Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm. → Report coverage: Report 3/4 measured, retaining the missing record separately.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



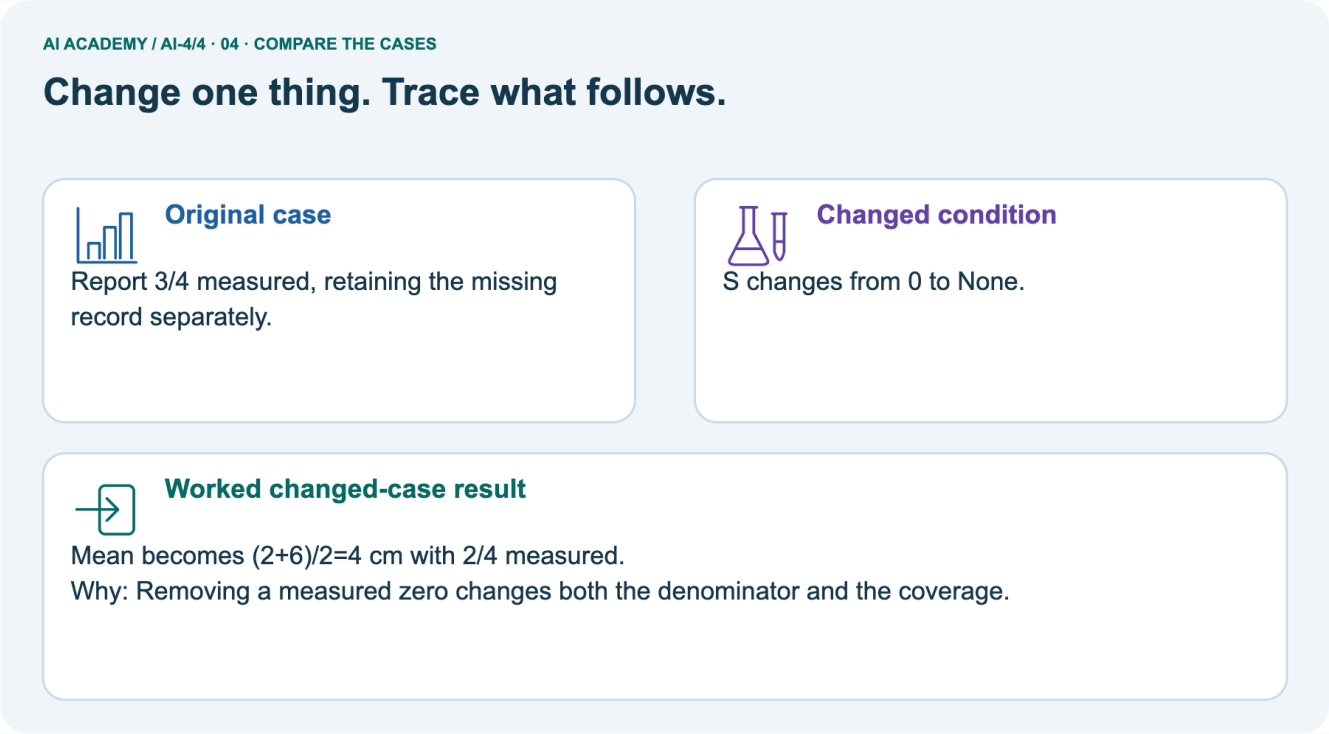
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report 3/4 measured, retaining the missing record separately. Changed condition: S changes from 0 to None. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

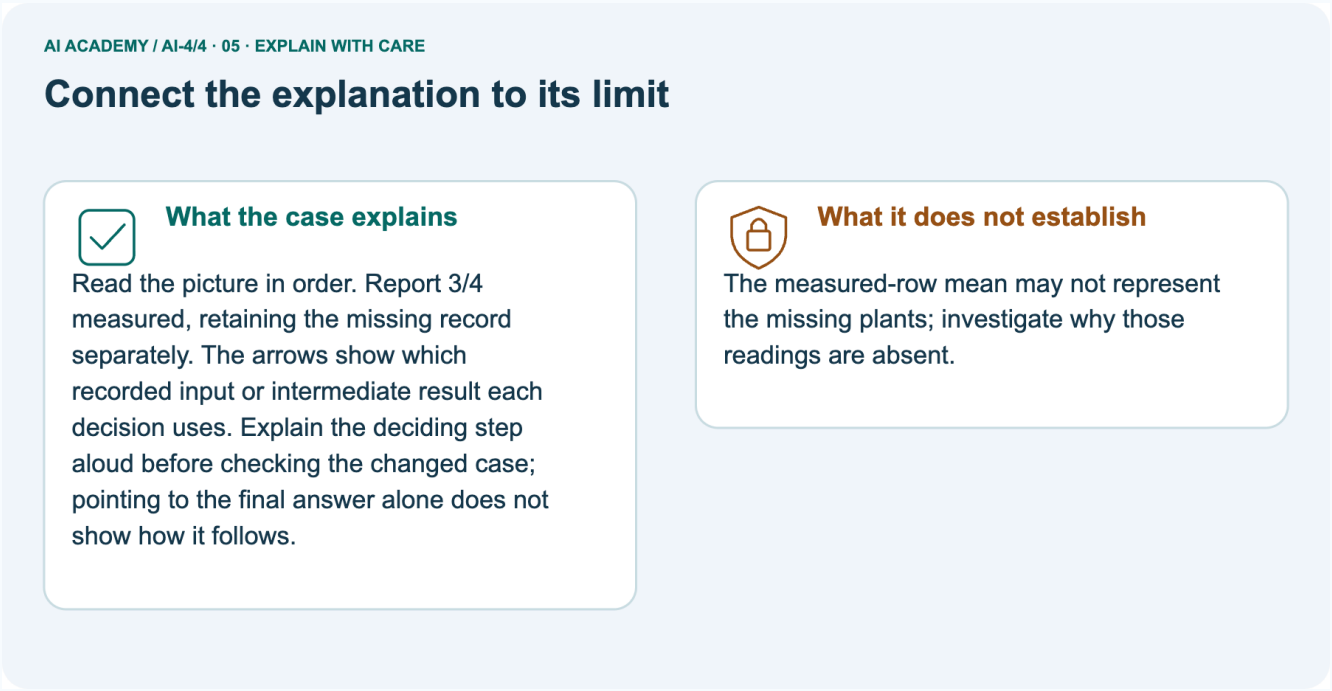
Reason: Removing a measured zero changes both the denominator and the coverage.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Removing a measured zero changes both the denominator and the coverage.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

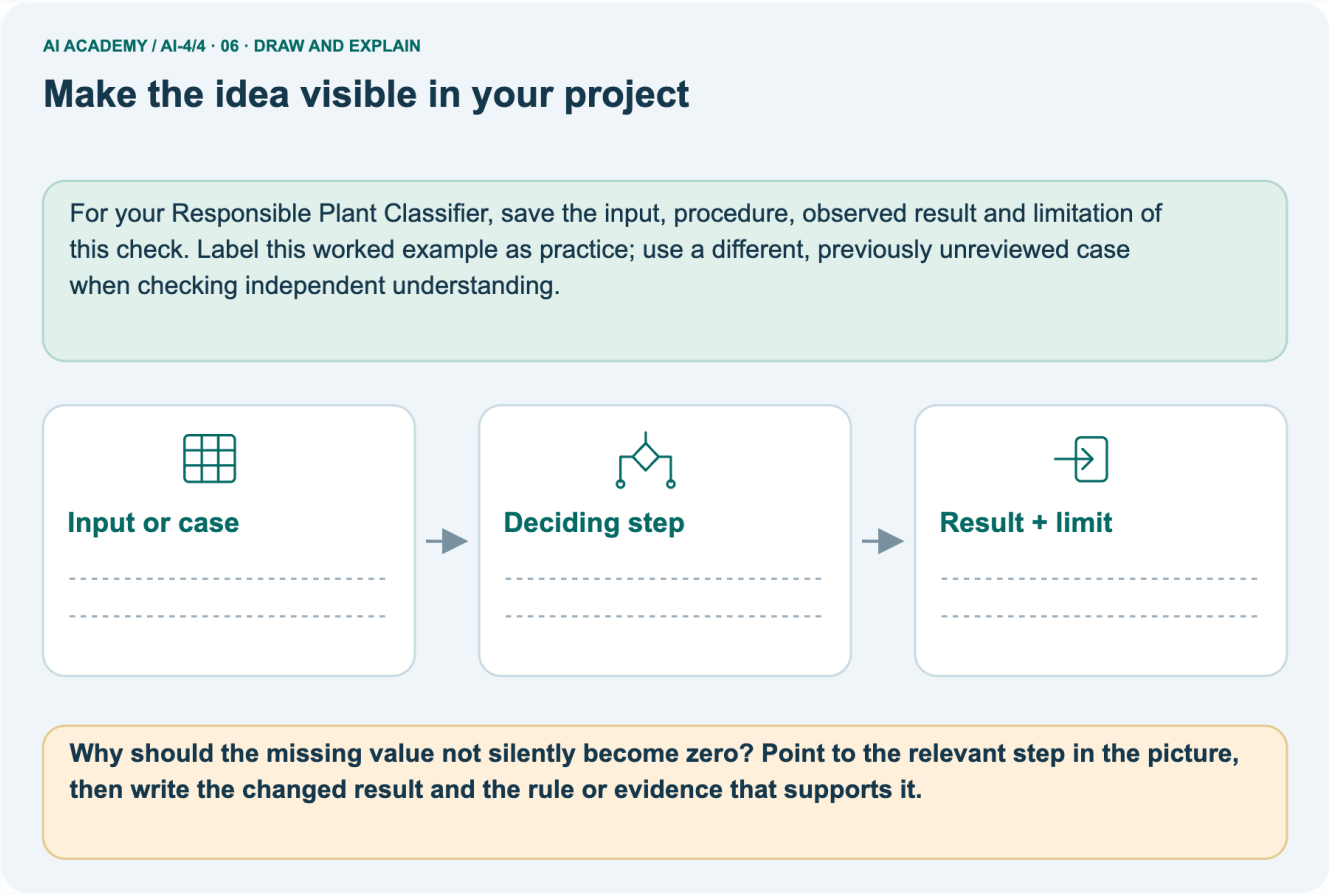
Limit: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Zero is a real recorded measurement here. Replacing missing with zero would invent **evidence** and lower the mean without justification. Changed-case check: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Removing a measured zero changes both the denominator and the coverage.

▼ **Earlier wording, alternatives and picture transcripts**

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Python and Table Readiness: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.